Bicol University POLANGUI CAMPUS

Centro Occidental Polangui, Albay

IT112: WEB SYSTEMS and TECHNOLOGIE'S

2nd Semester, AY: 2025-2026

Phase 3: Final Group Project

BentaBoard

"From bawat benta to tunay na kita—track it right."

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IT112 Professor

Introduction

After finishing the backend and database setup in Phase 2, our group, "**PeNiHAMiNo**," moved on to "**Phase 3: Frontend Development**". This part of the project is where we turn our technical plans, like the ERD and Use Case diagrams, into actual web pages that a user can see and click on.

The goal of **BentaBoard** is to help small business owners move from "**bawat benta**" (every sale) to "**tunay na kita**" (actual profit). To make this happen, we needed a user interface (UI) that is simple, clean, and easy to use, even for people who aren't good in technology. We used HTML5, CSS3, and Bootstrap 5 to build the site. We chose Bootstrap because it makes the website "responsive," meaning it will look good and work correctly whether the user is opening it on a laptop, a tablet, or a smartphone.

In this phase, we focused on making sure that every page we built has a purpose. We didn't just make it look pretty; we made sure that every input field and button corresponds to the data we planned in our database. This ensures that when we connect the frontend to the backend in the next phase, everything will fit together perfectly.

Objectives

<u>At the end of this activity, we were able to:</u>

1. Translate project design documents (ERD, use case, architecture) into responsive HTML/CSS/Bootstrap interfaces.
2. Create modular, consistent, and functional frontend components that match the system's data model.
3. Prepare and connect pages for later backend/API interaction (planned in Phase 3).
4. Utilize Bootstrap components for layout, structure, and responsiveness.
5. Collaboratively commit their work to GitHub following a structured frontend workflow.

Frontend Folder Structure

```
/frontend
├── admin-dashboard.html
├── admin-login.html
├── admin-register.html
├── admin-reports.html
├── admin-settings.html
├── admin-users.html
├── analytics.html
├── dashboard.html
├── expenses.html
├── index.html
├── inventory.html
├── login.html
├── purchases.html
├── register.html
├── reports.html
├── role-select.html
├── saleslogs.html
```

```
├── settings.html
├── weather.html
├── /css/
└── style.css
├── /js/
└── admin-script.js
└── script.js
├── /img/
└── bentaboard.png
```

I. Overview of Frontend Setup

Our frontend system is developed using HTML, CSS, and Bootstrap 5 to provide a modern and responsive user experience. To maintain consistency with our Phase 2 backend organization, we followed a modular folder structure.

The frontend is organized to separate the structure (HTML), the styling (CSS), and the logic (JS). This ensures that the project is easy to maintain and that multiple team members can work on different pages without causing code conflicts.

The frontend folder is structured as follows:

/frontend – The root directory containing all main HTML pages (e.g., index.html, dashboard.html).

/css – Contains custom CSS files to handle the specific branding and colors of BentaBoard.

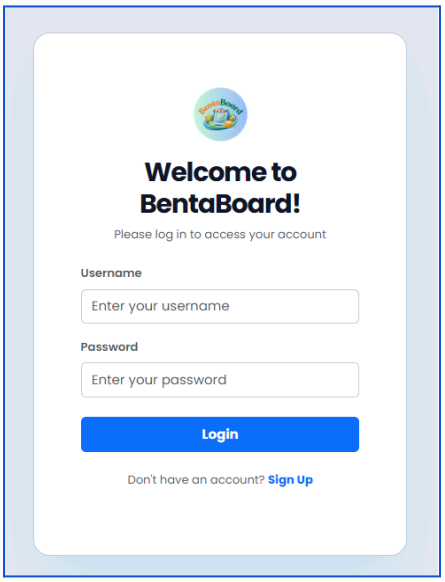
/js – Contains JavaScript files used for form validation and the future API integration scripts.

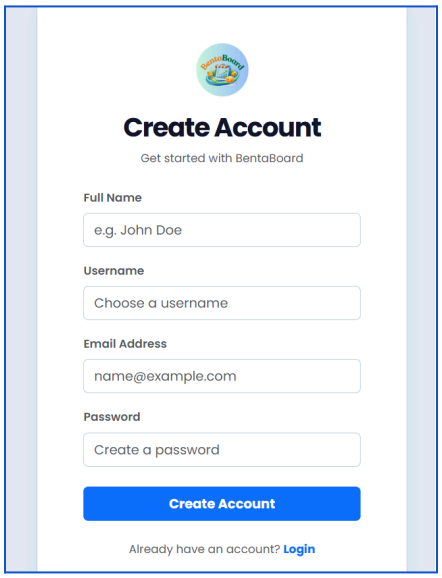
/img – Stores all optimized icons, logos, and images used across the interface.

This setup ensures that the frontend is scalable, making it easy to add new pages or features as the project grows.

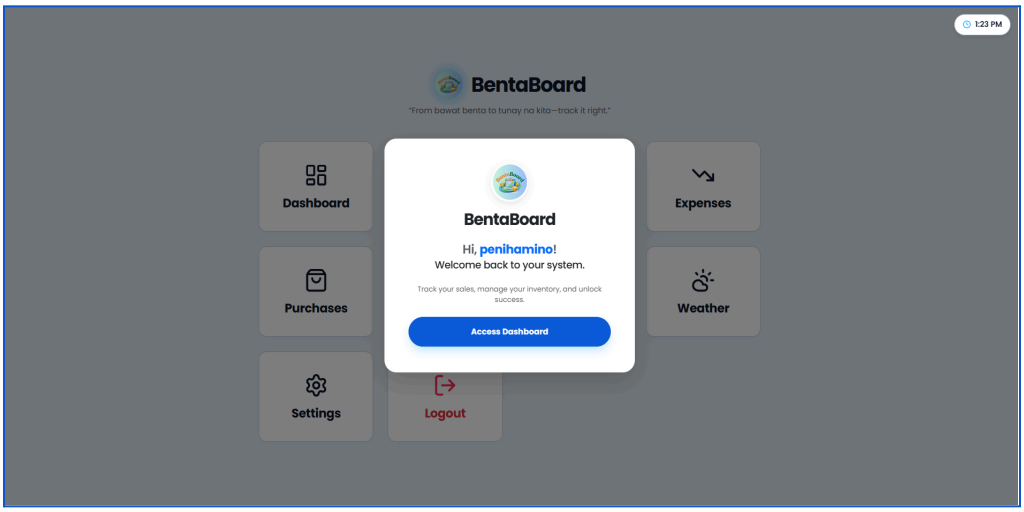
II. Screenshots of ALL working frontend pages designated only for SELLERS.

- **Create/Login & Register/Sign Up**

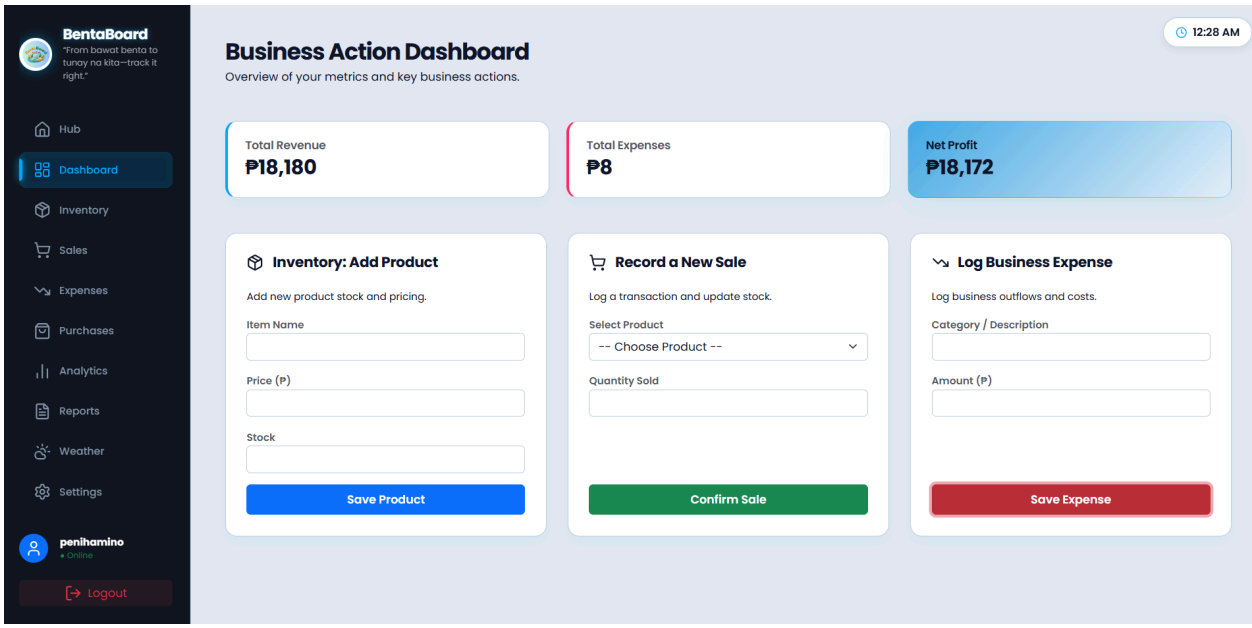




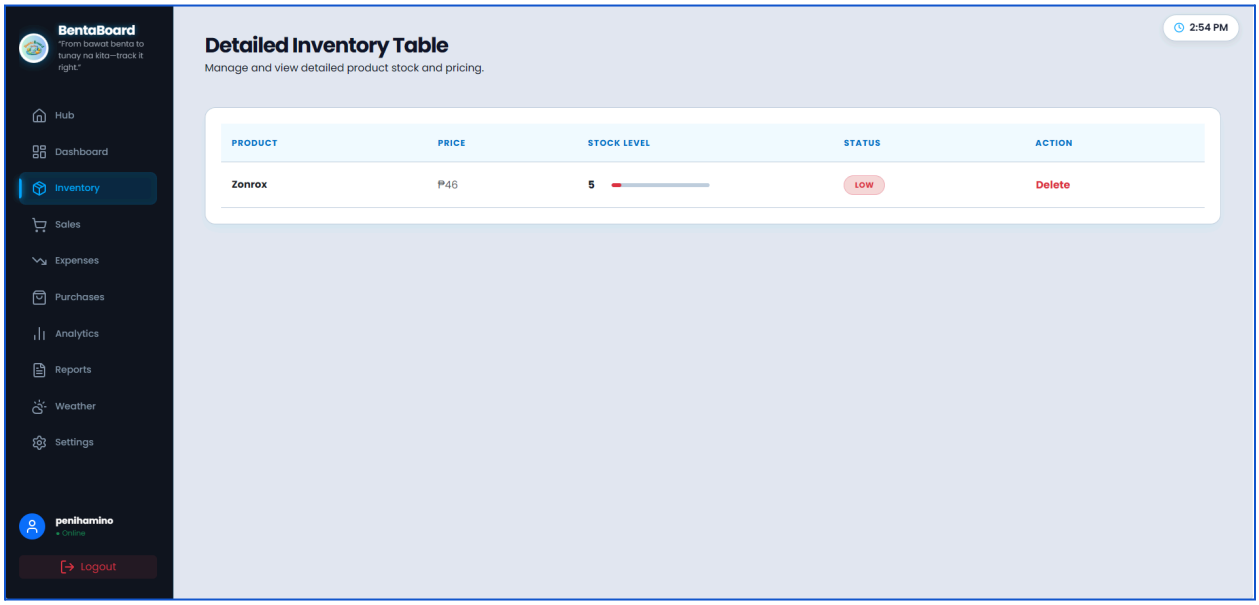
- **Hub**



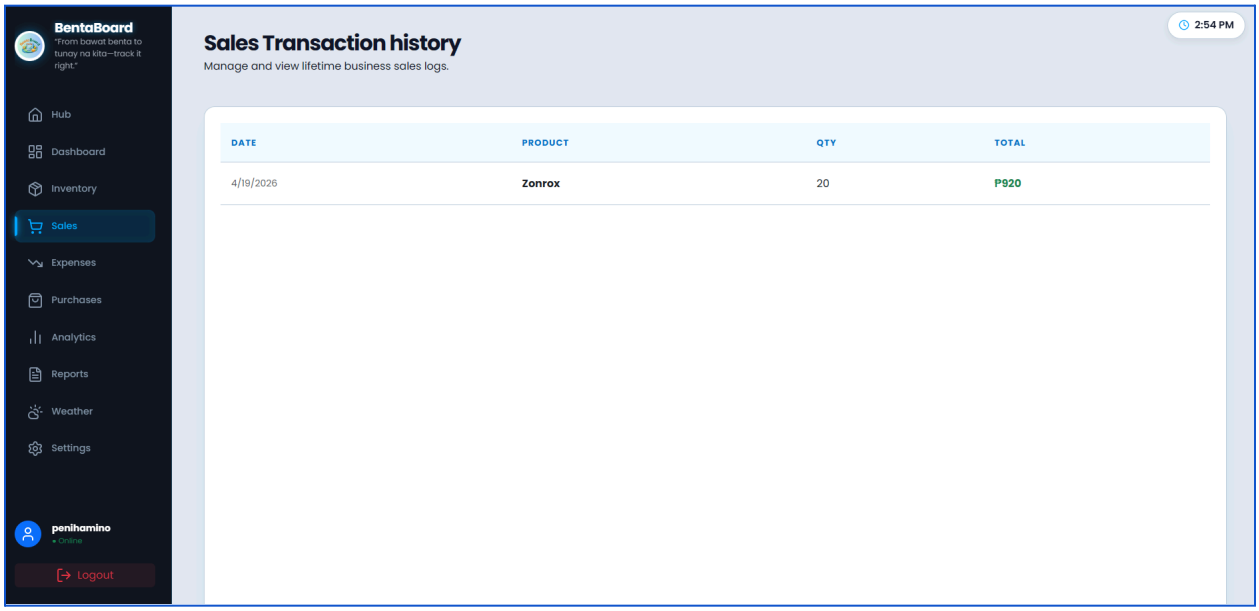
- **Dashboard**



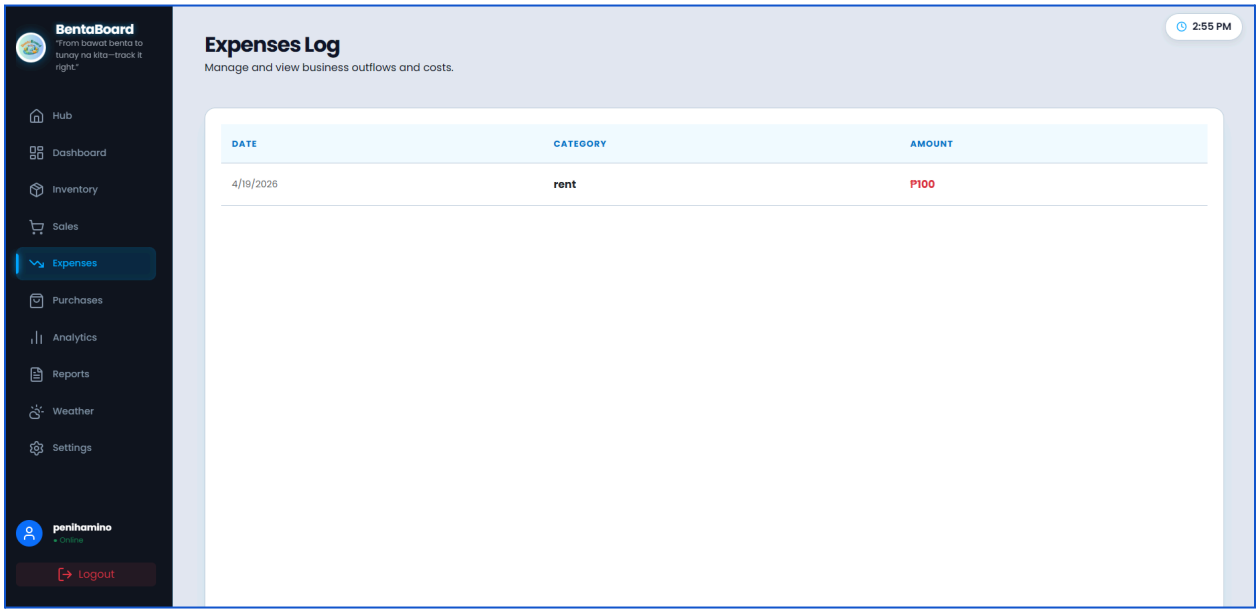
- **Inventory**



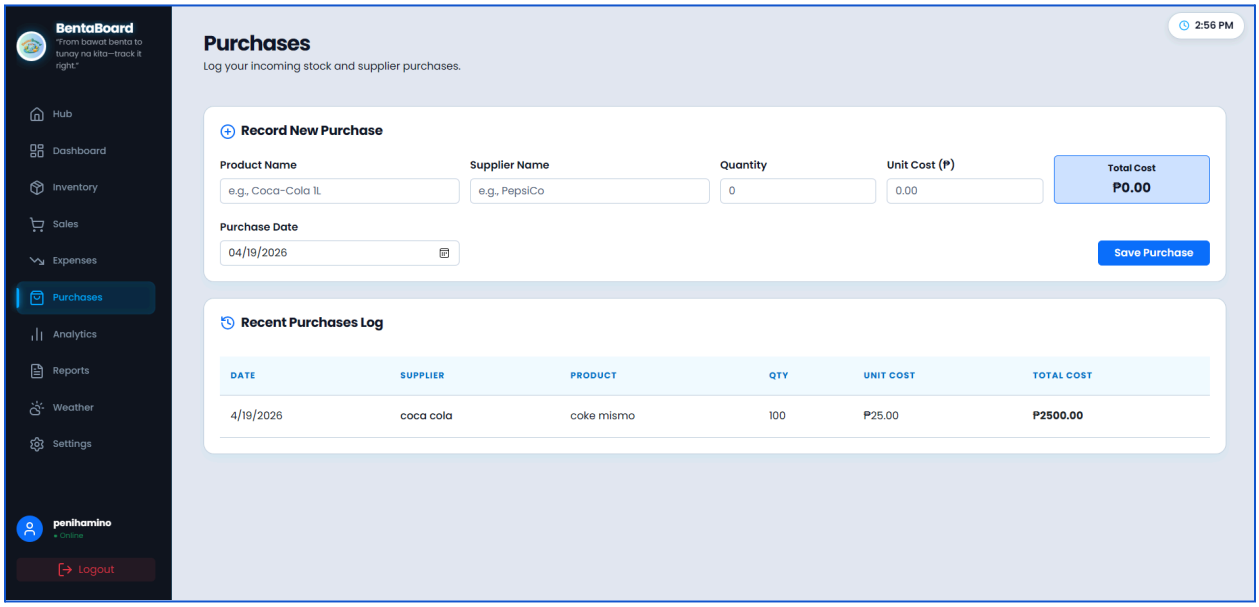
- **Sales**



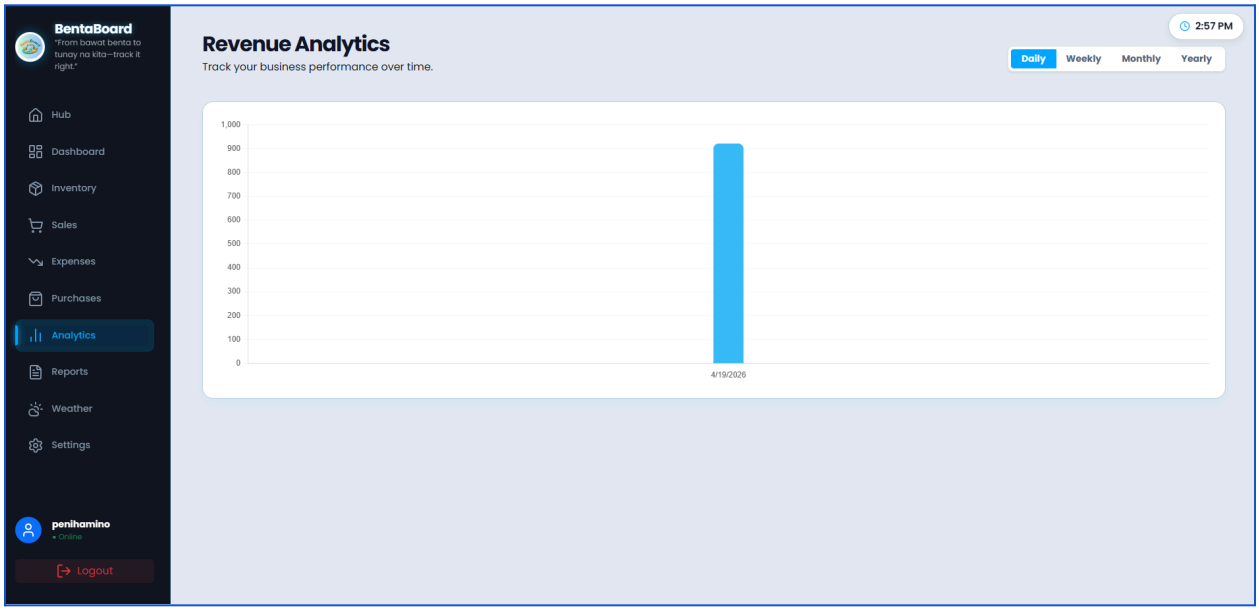
- **Expenses**



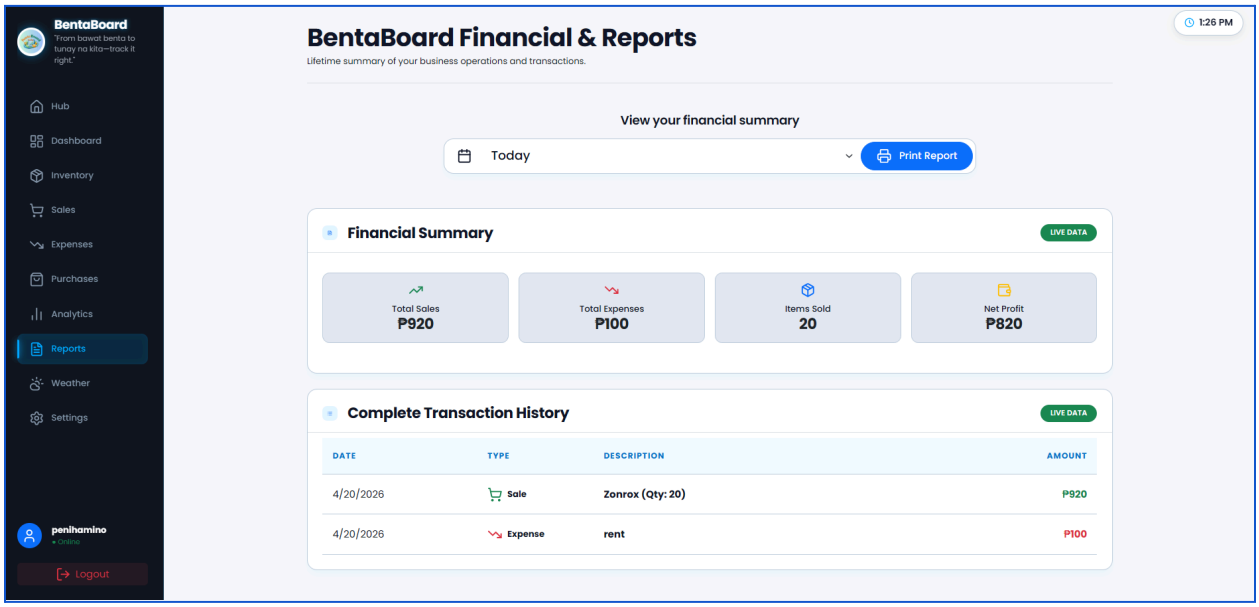
- **Purchases**



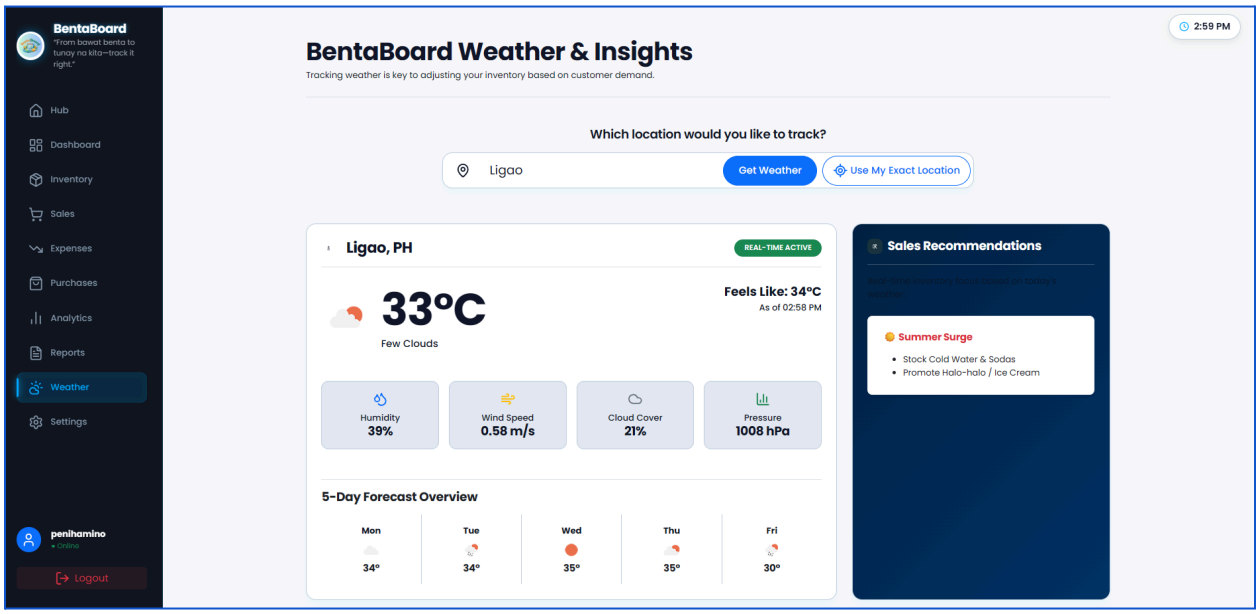
- **Analytics**



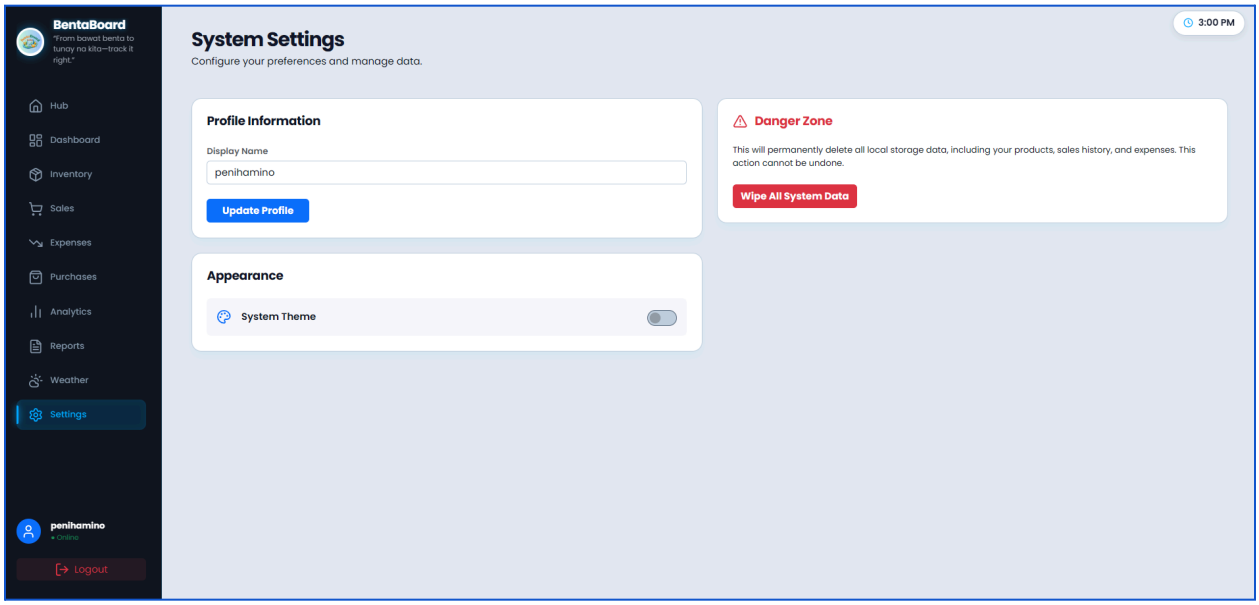
- **Reports**



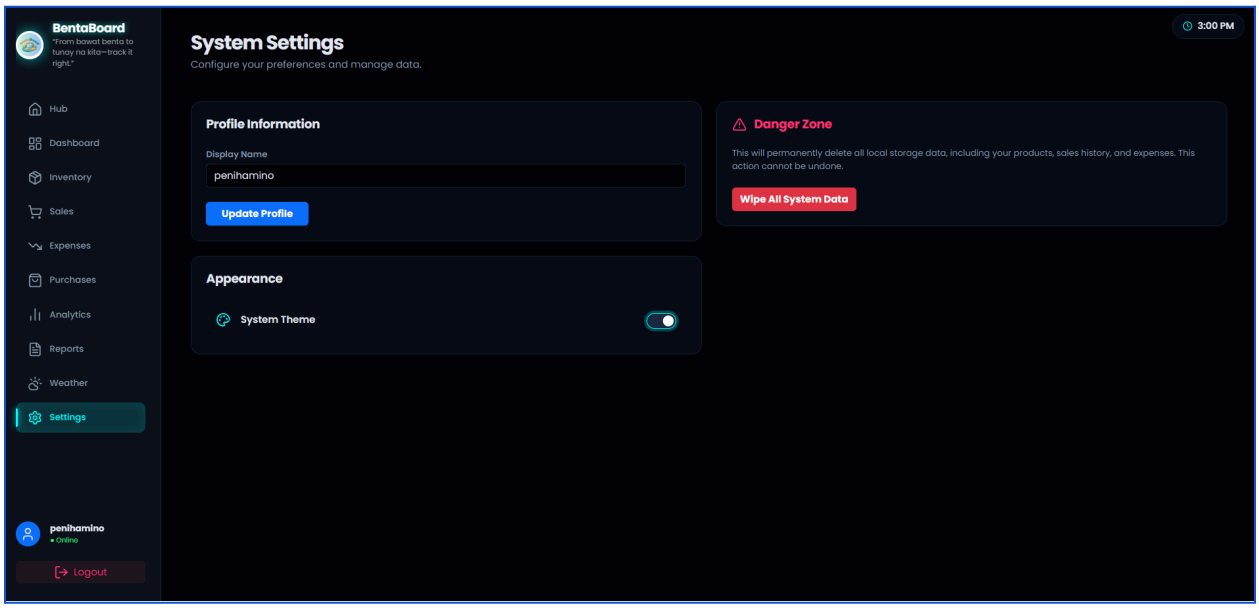
- **Weather**



- **Settings**



- **Darkmode**

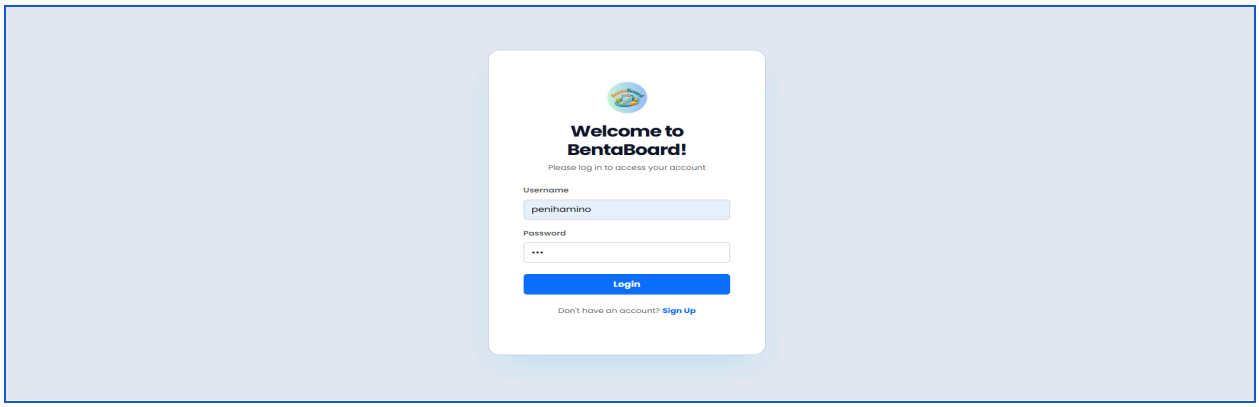


III. Annotation showing which feature/use case each page covers

We carefully followed our initial project proposal to ensure every feature we promised was included in the design. We organized our files into a /frontend folder to keep the project neat.

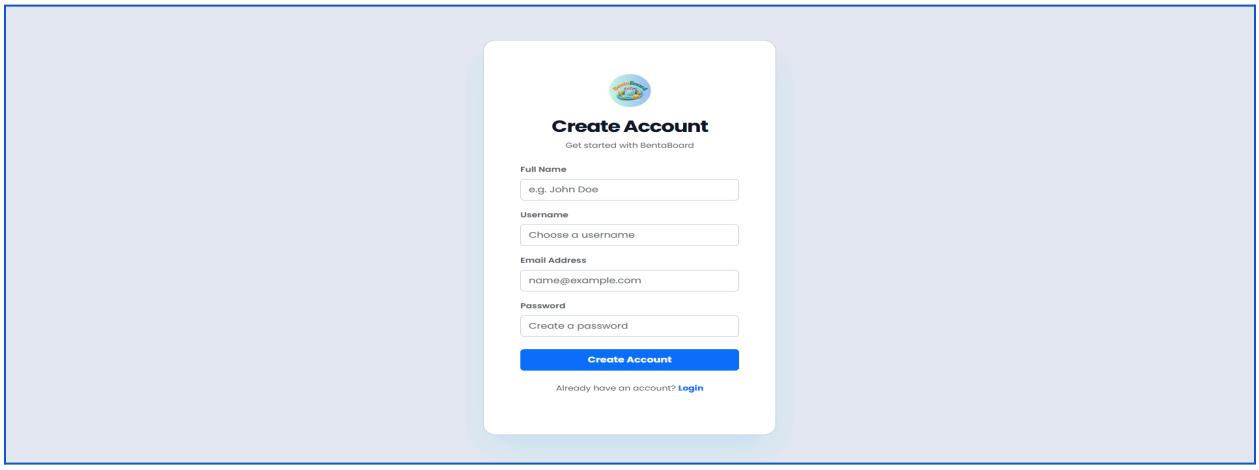
1. User Authentication (Login & Registration)

Based on our Use Case diagram, every user must be authenticated. We created a login and registration page with fields that match our User schema in the database.



- This page allows users to create an account and log in securely to access their business dashboard.

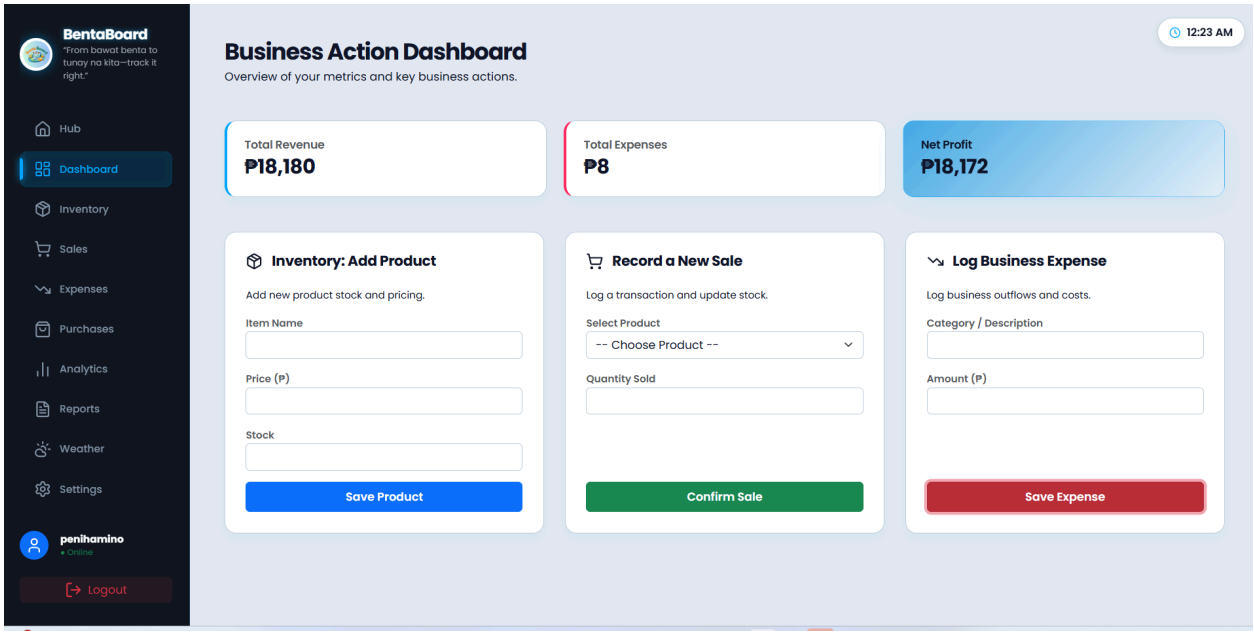
2. User Authentication (Sign Up Form)



- The sign-up form serves as the entry point to BentaBoard. It collects new user details like name, username, email, and password to create an account.

3. Main Dashboard

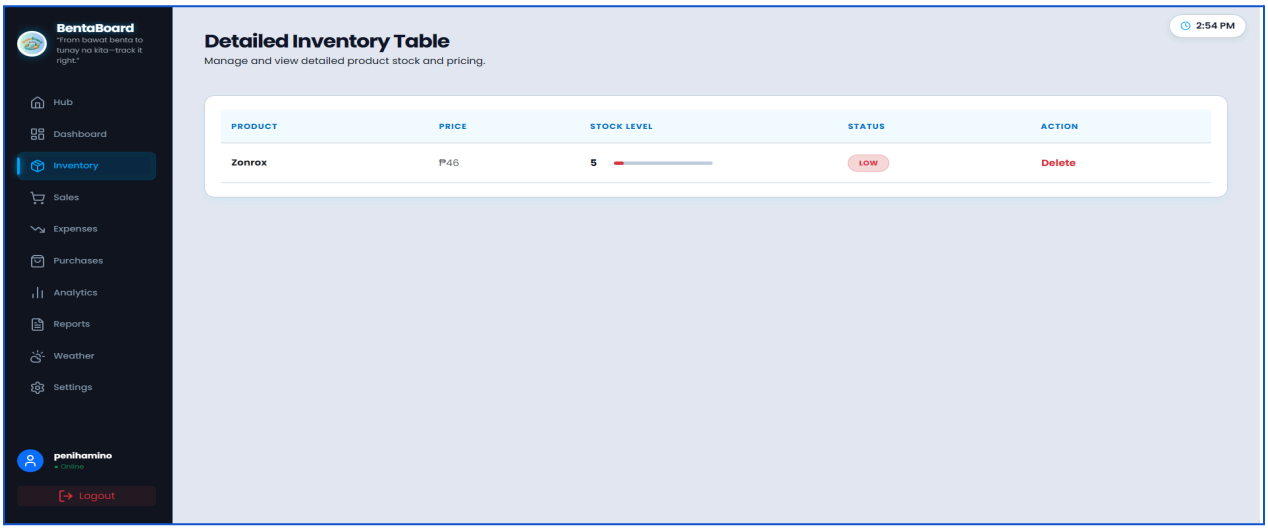
The dashboard serves as the central hub. It provides a quick overview of the business's financial health, showing total sales and profits.



- The dashboard gives the seller a snapshot of their daily and monthly earnings.

4. Detailed Inventory

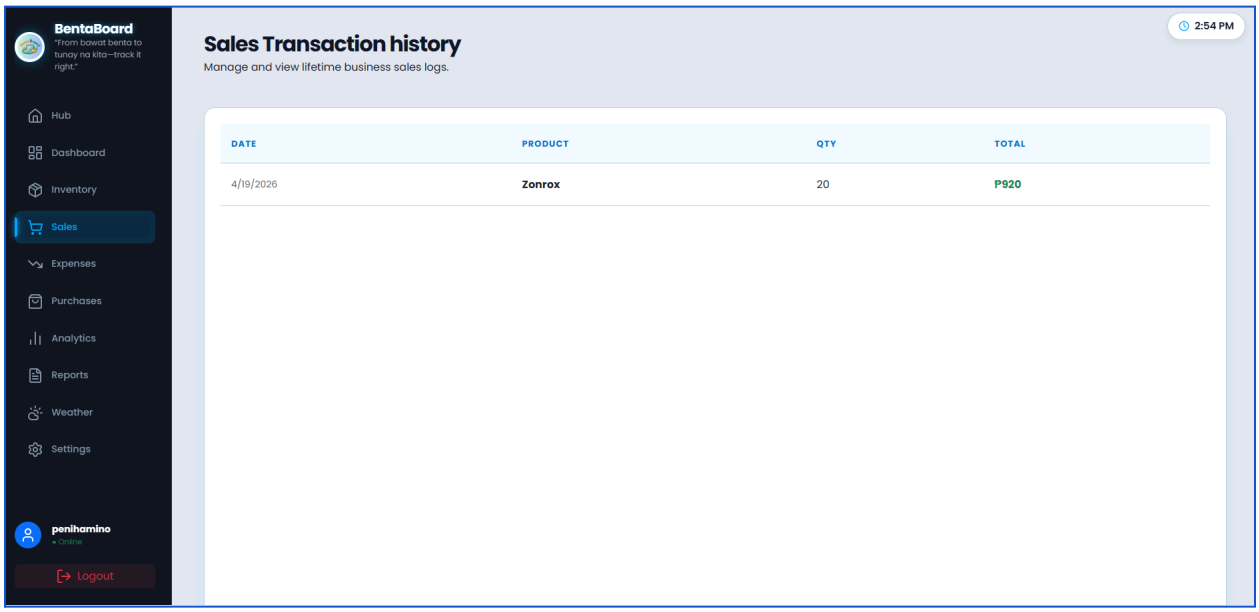
Following our ERD, we built a page where sellers can see the status of the products and delete their products. This page includes fields for product names, prices, stock levels, and status.



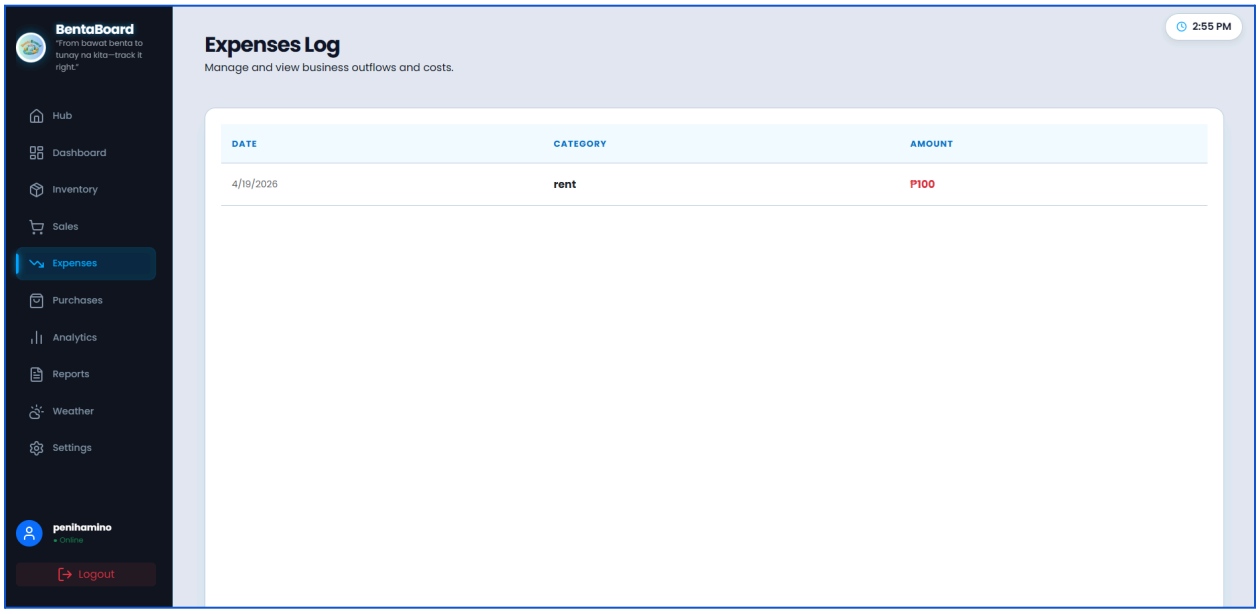
- This page handles the "Inventory Detailed" use case, allowing users to keep their stock list updated.

5. Sales and Expense Tracking

To fulfill the main purpose of BentaBoard, we created dedicated forms for recording daily sales and operational expenses.

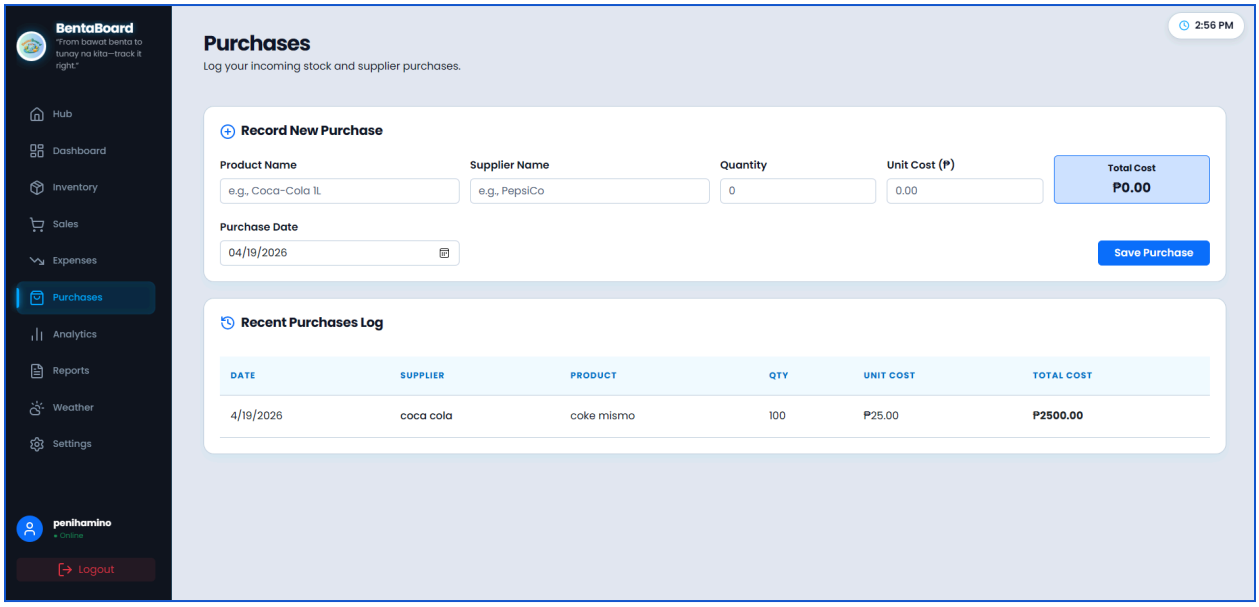


- This is where sellers log their daily transactions and quantities sold.



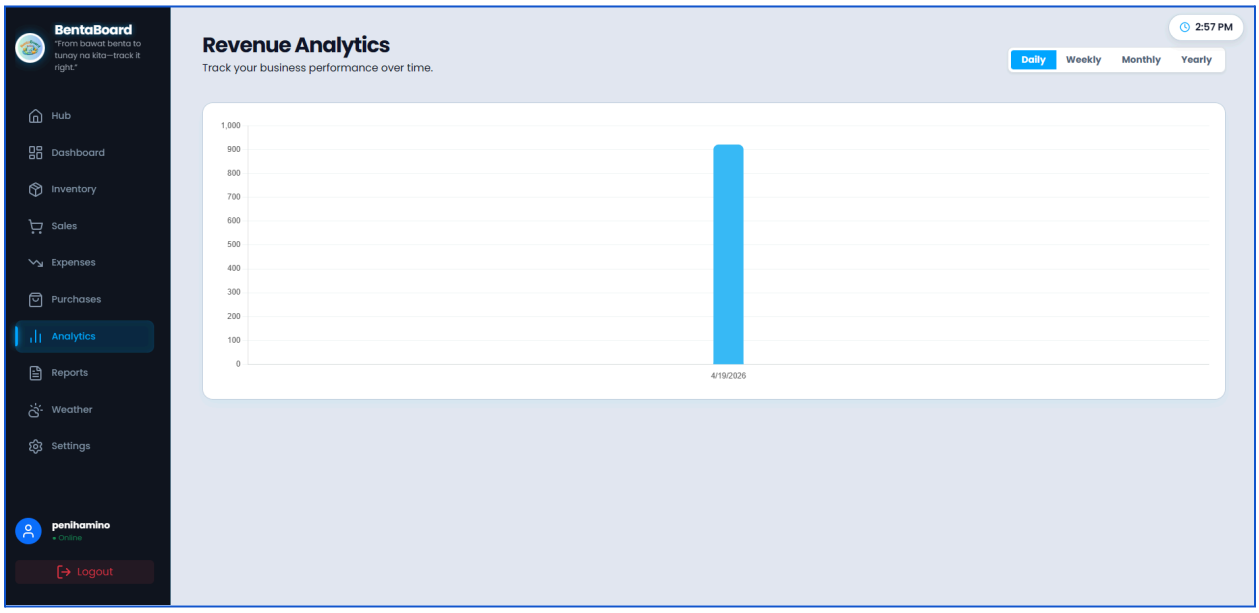
- This page allows users to track costs like rent, electricity, and transportation.

6. Purchases



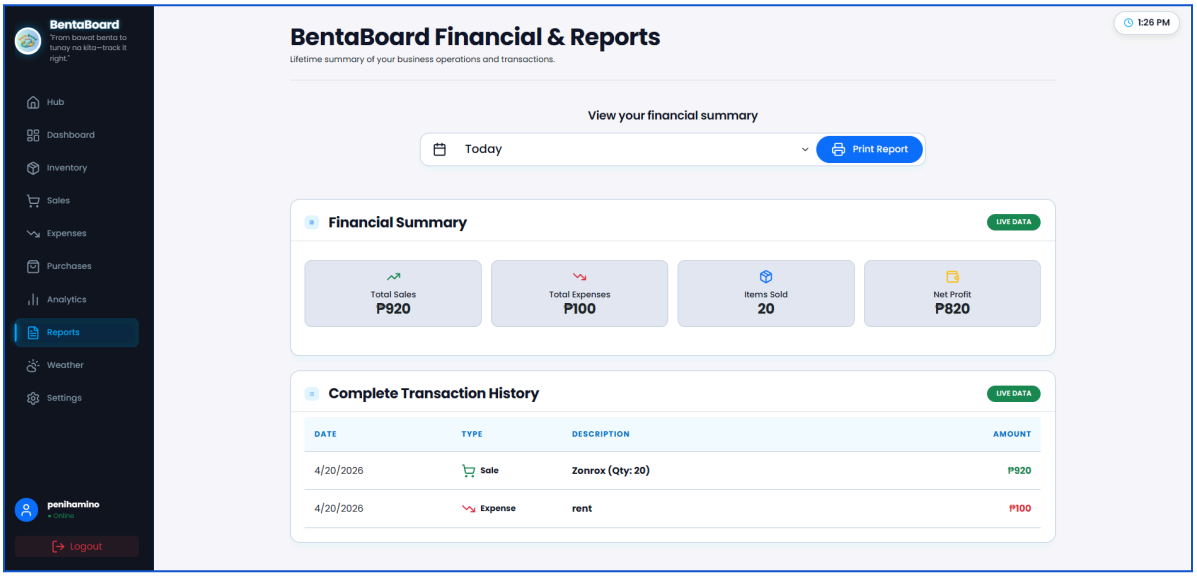
7. Reports and Analytics

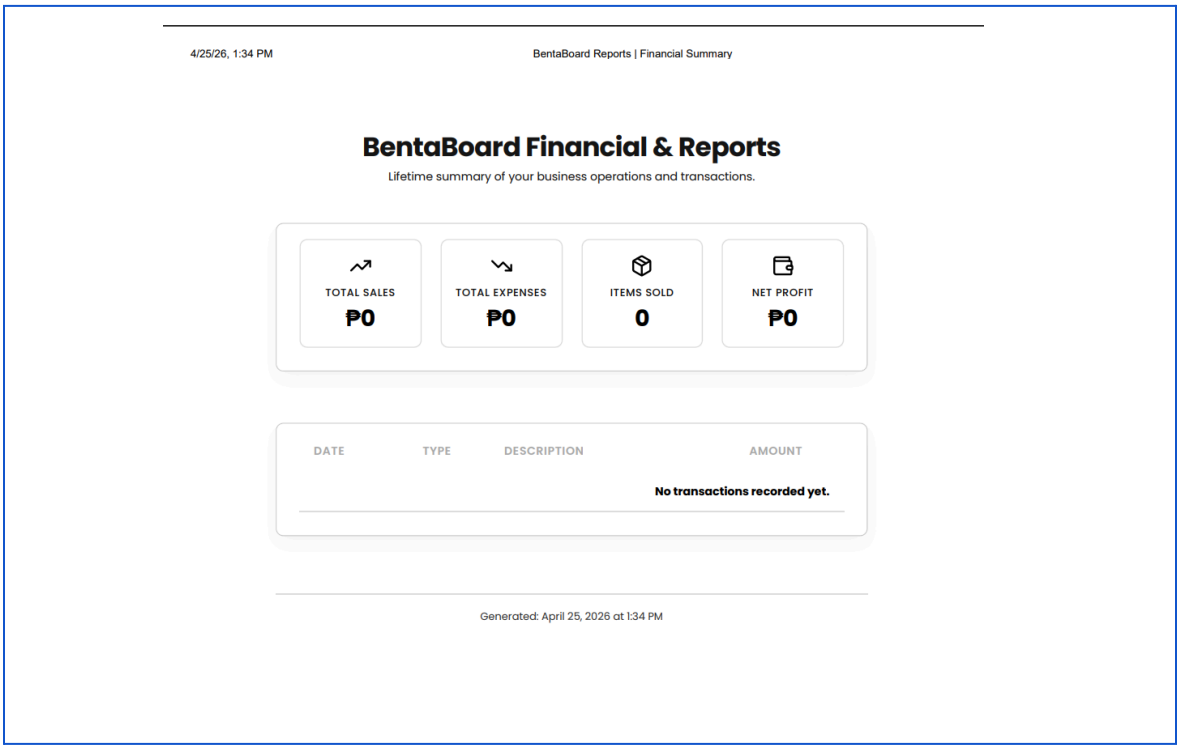
We designed a reports page that will eventually display the automated profit calculations (Total Sales - Expenses/Purchases).



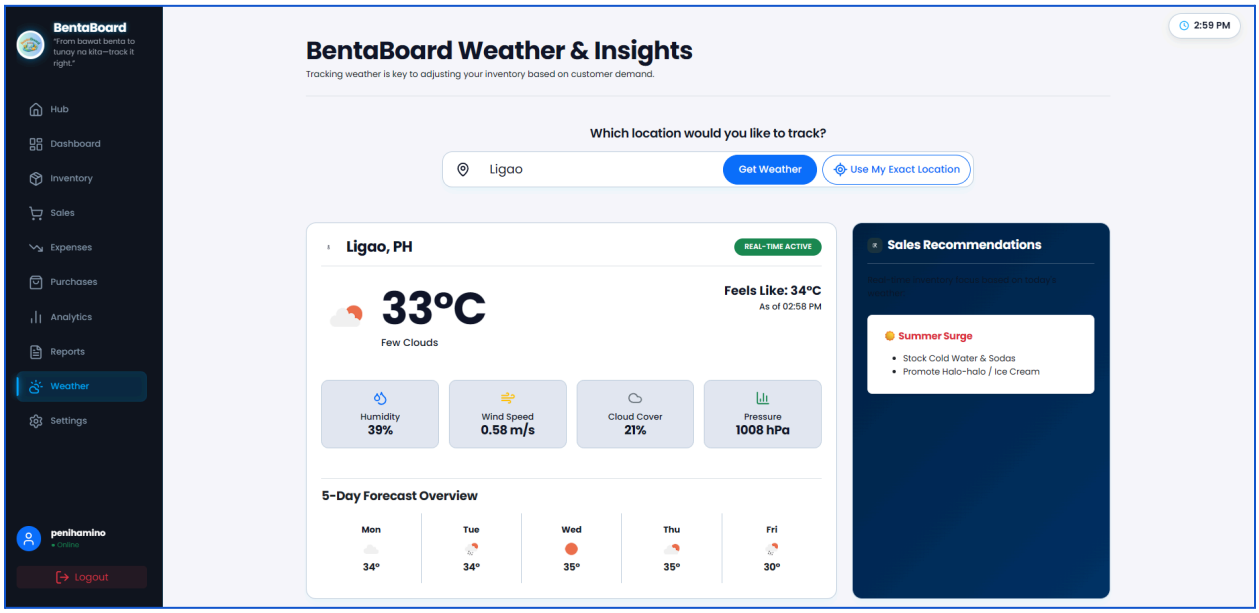
- This page is designed to show the final profit reports on a daily, monthly, and yearly basis.

- **Report**



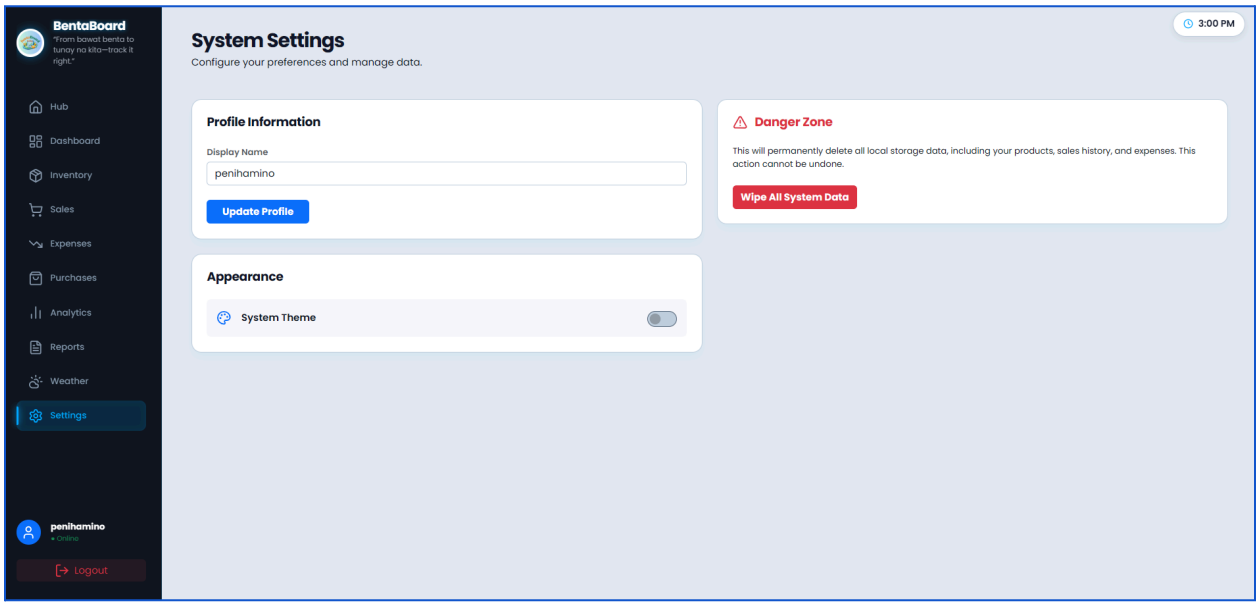


7. Weather



- We added a special page that shows the current weather. We did this because weather often affects how many customers visit a store. By having this integrated, the user can check the weather and plan their inventory or sales targets for the day.

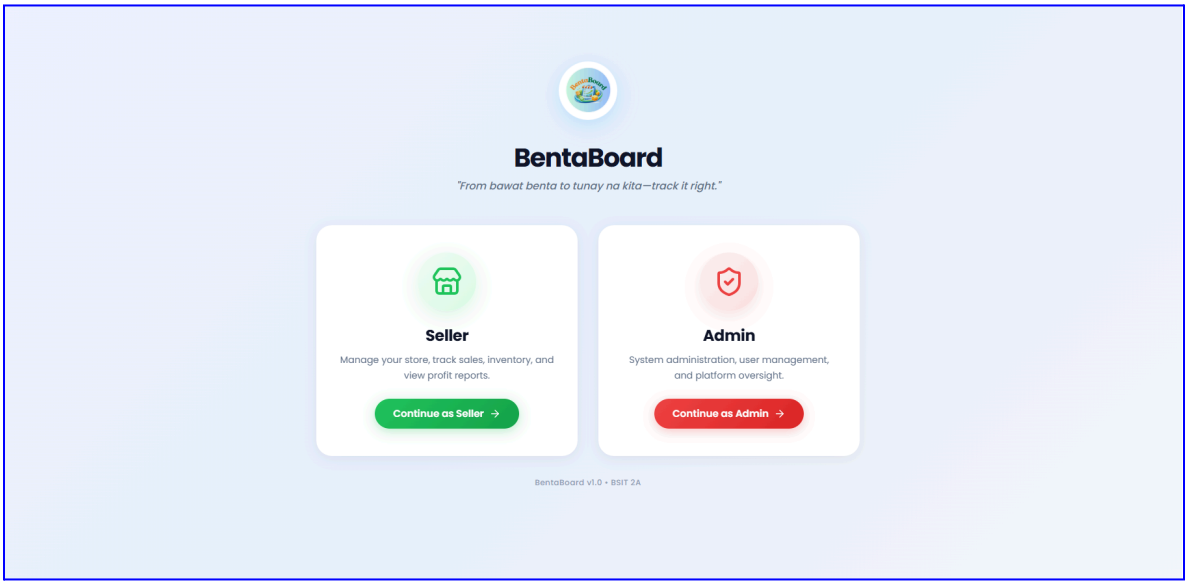
8. Settings



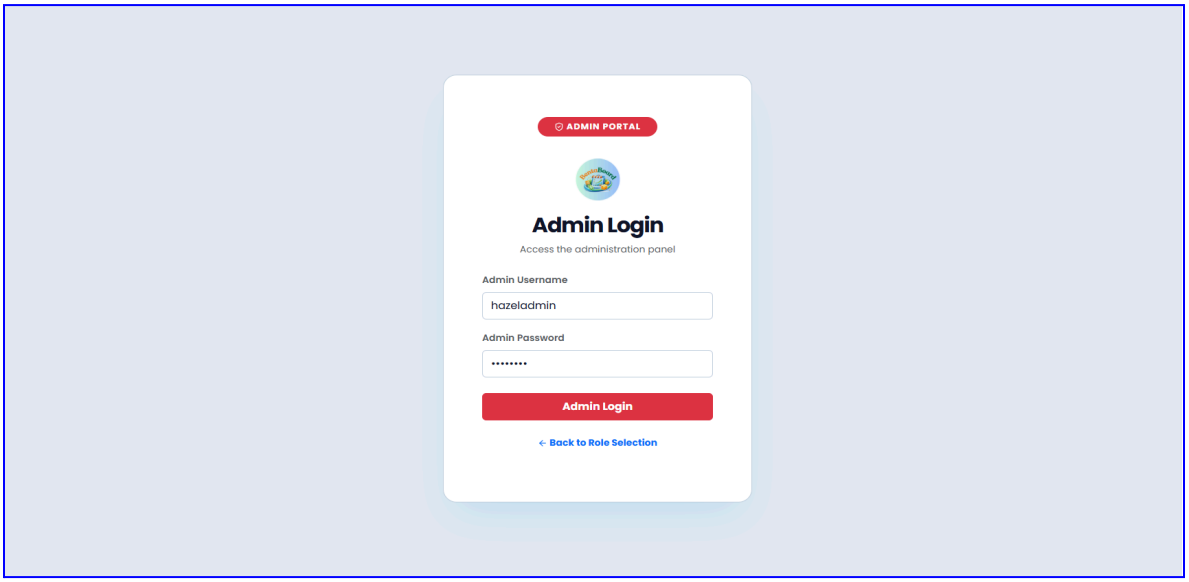
- We made a settings page. This is where the user can change their profile information or update their password. It's a simple page with a few input fields, making sure the user has full control over their account.

II. Screenshots of **ALL working frontend pages** and **annotation** showing which feature/use case each page covers that works only for **ADMIN.**

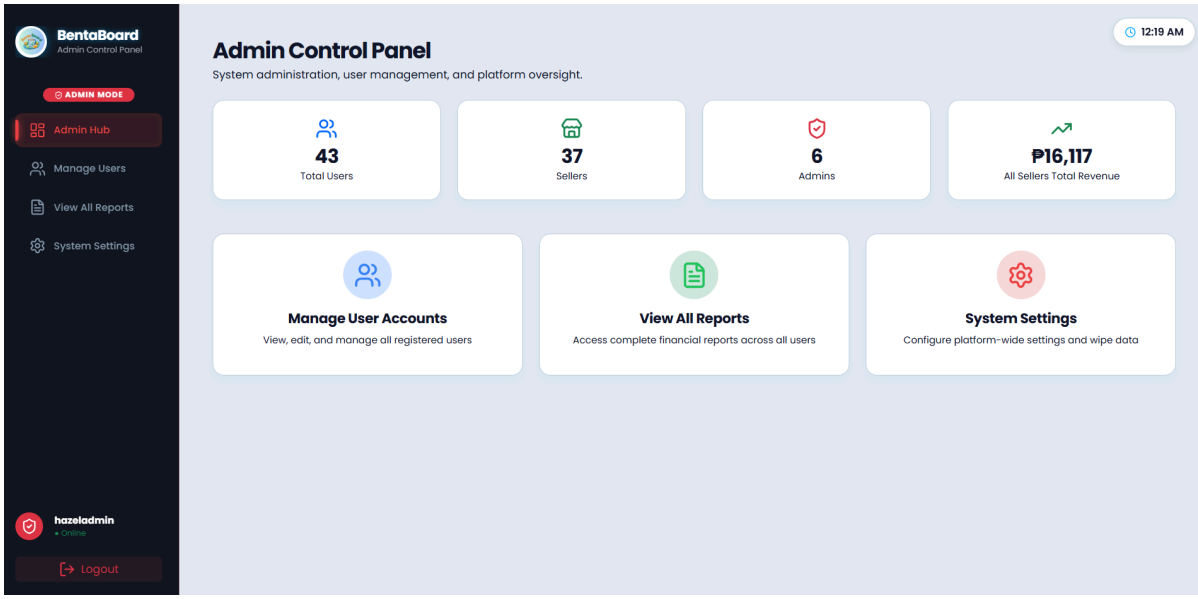
- First working frontend right after opening BentaBoard. It contains separate system view that is designated only for **sellers** and **admins**.



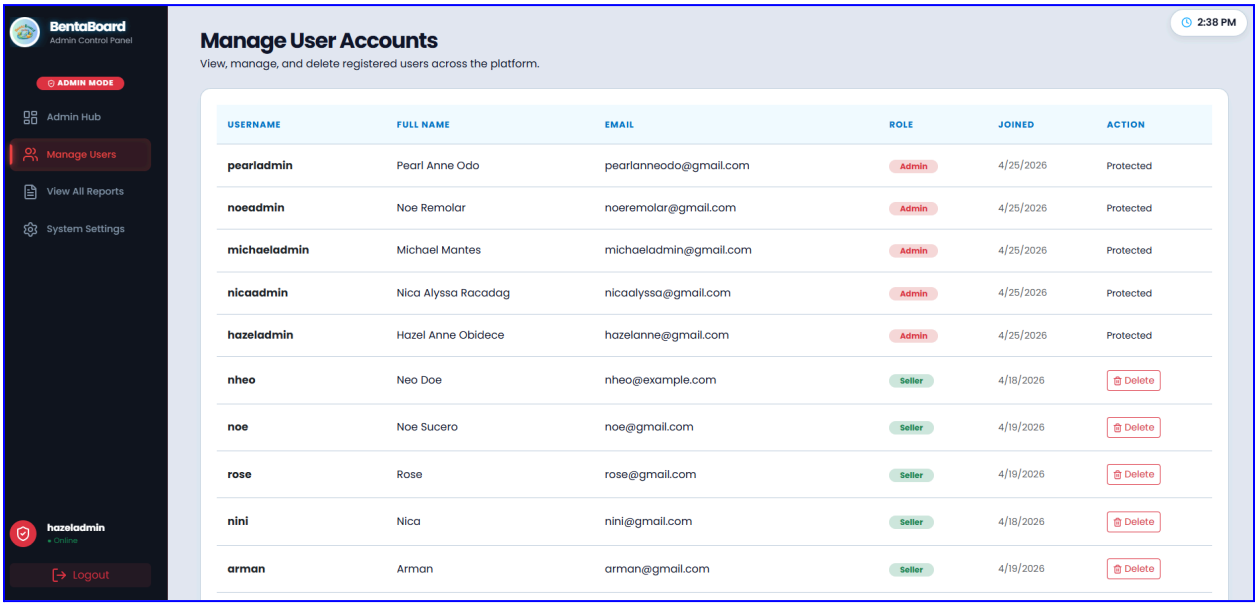
- **Admin Portal: Login page for admins only**



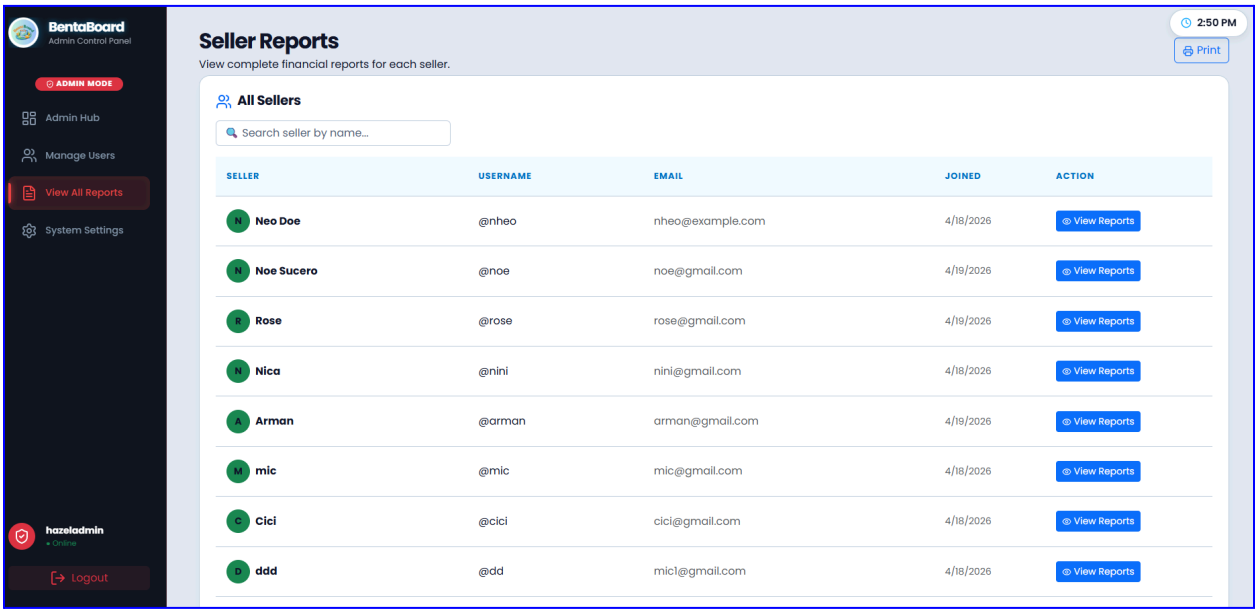
- **Admin Hub:** This is where the admin hub will take place. A full overview of **Total users, sellers, admins of BentaBoard as well as the Total Revenue of all Sellers.** The sidebar contains “Manage Users”, “View all reports of sellers”, and full “System Settings” of BentaBoard that only Admins could access.



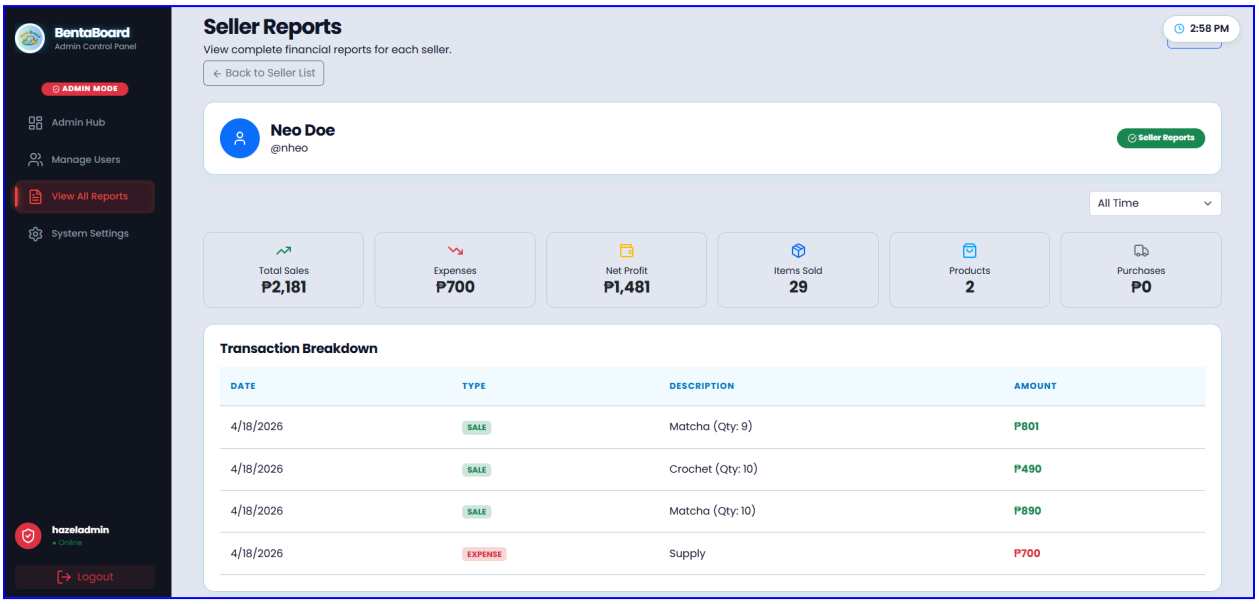
- **Manage Users:** Manage Users Hub allow admins to see the details of all BentaBoard users. These include their *Username, Full Name, Email, Roles, and Date Joined.*



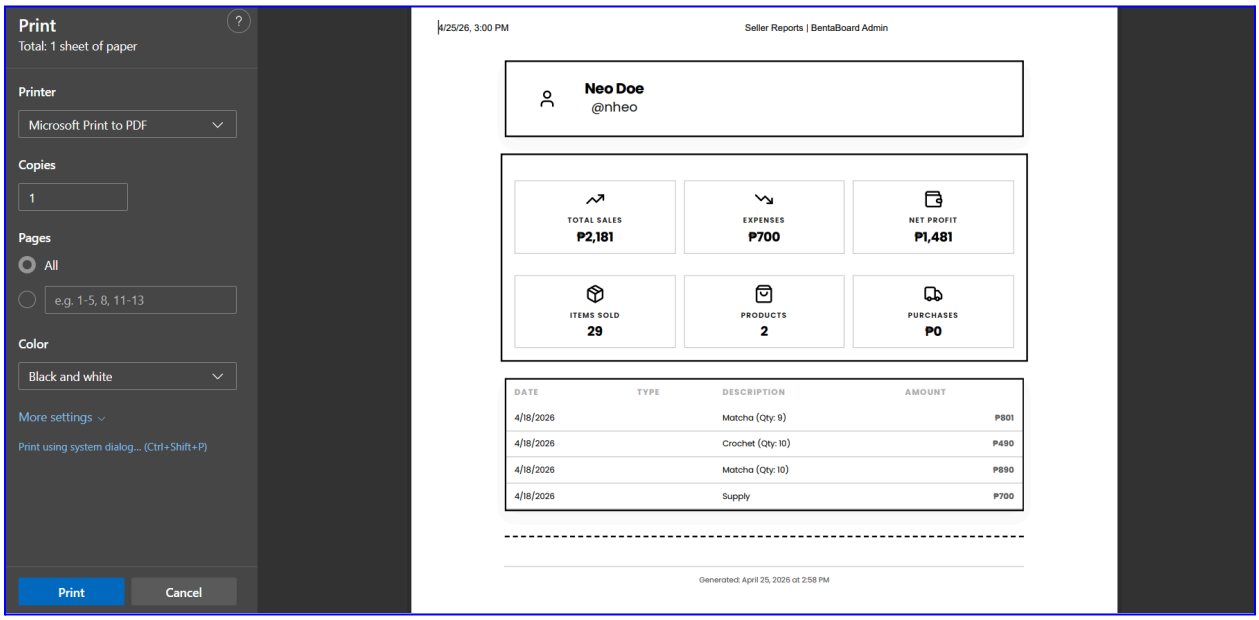
- **View all Reports:** Admins also have the ability to check each seller's activities through **Reports**.



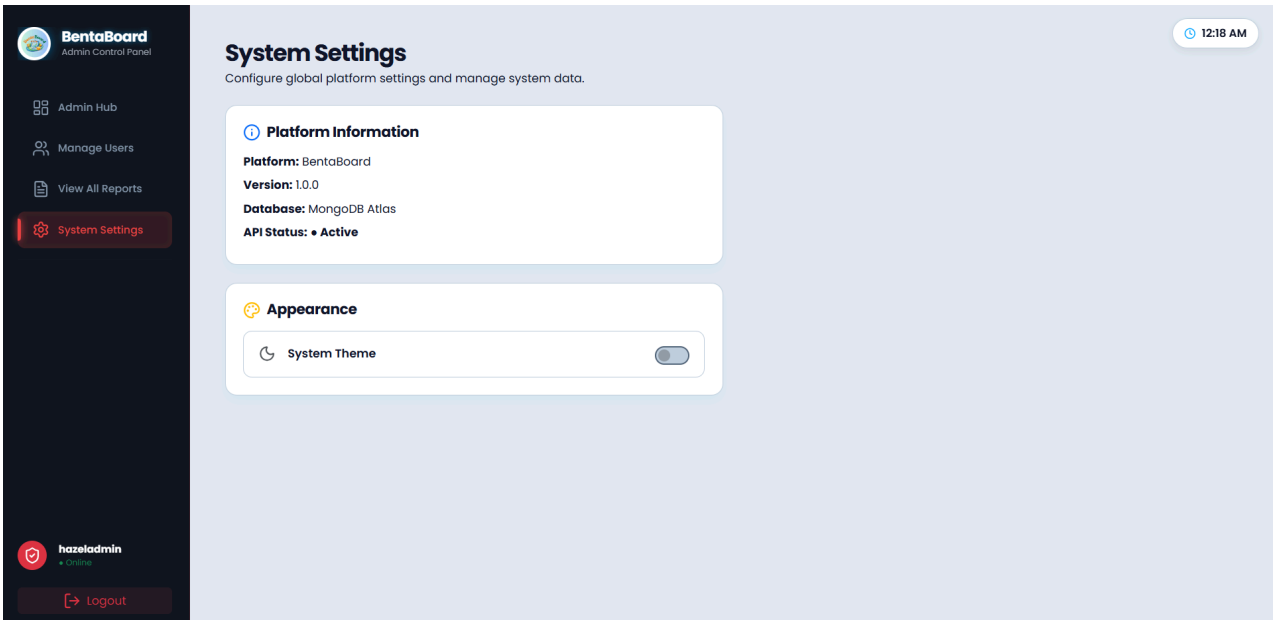
- Example reports of **Neo Doe:**



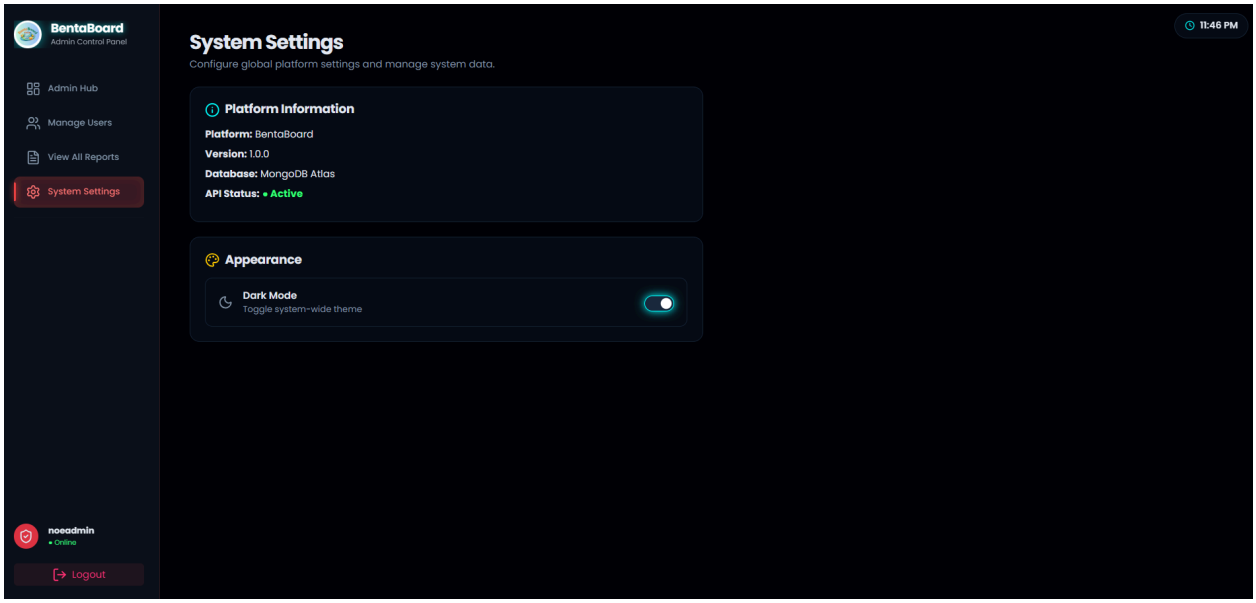
- **Example Reports** <u>print preview</u> of Neo Doe:

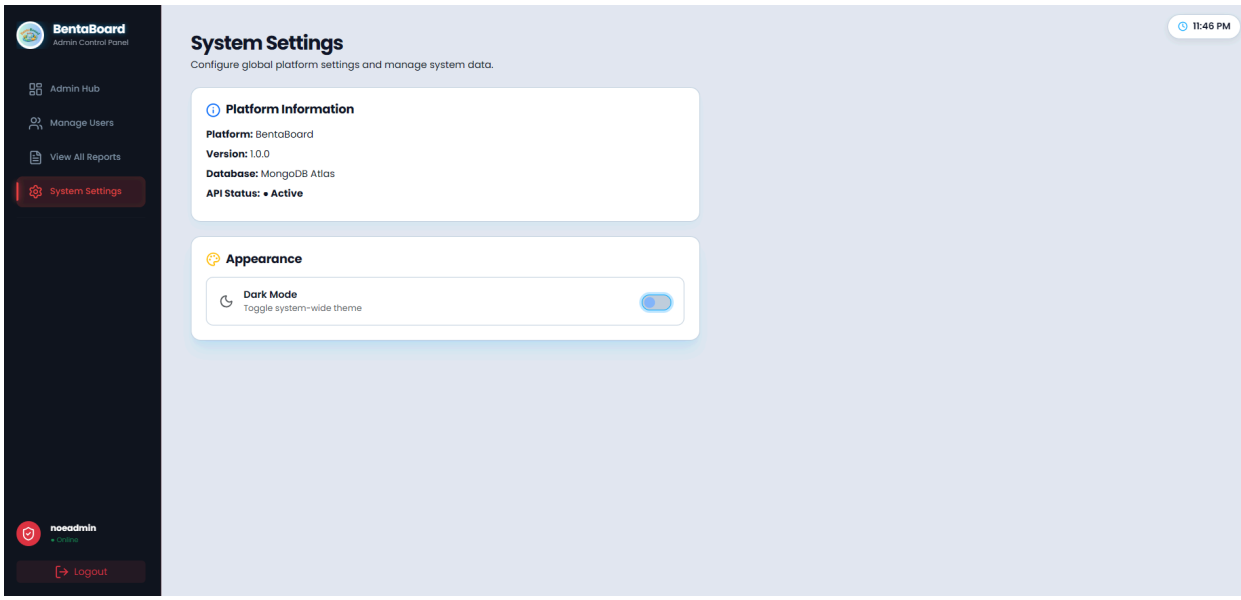


- **Admin System Settings:** Full overview of BentaBoard settings (for admin point of view). This tab contains *Platform information, and System Appearance*

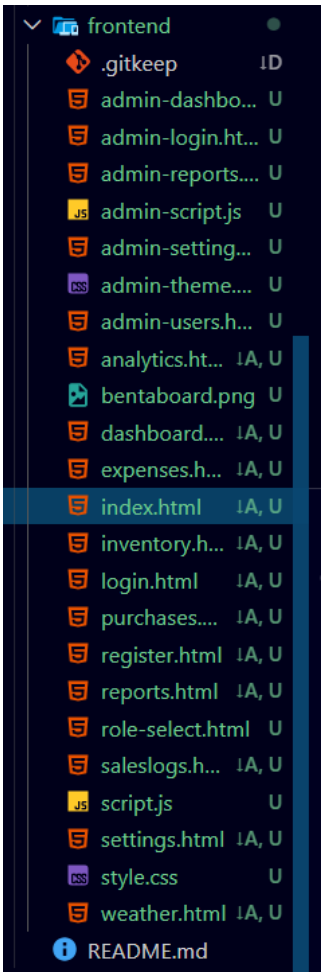


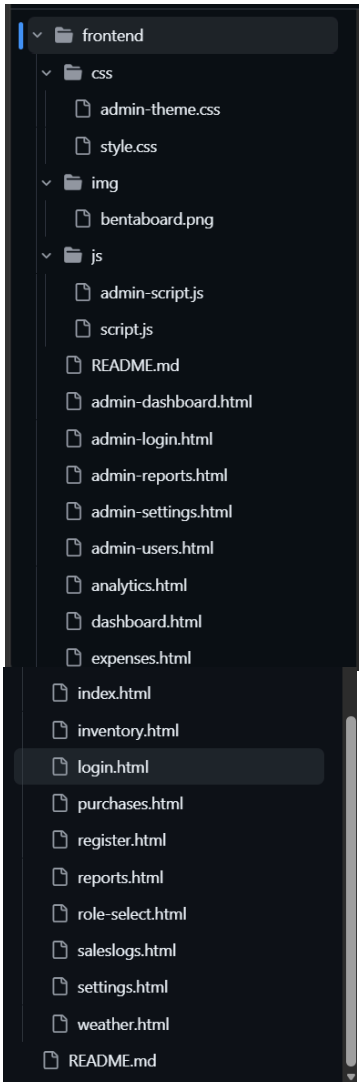
- When button toggled into **Dark Mode:**





IV. Screenshot of GitHub folder structure





V. Individual reflections from each group member (3–5 sentences)

Hazel Ann Obidece - Project Manager & Backend Developer

During this phase, I made sure that the pages we built actually matched our original proposal. It was challenging to ensure the flow of the website felt natural, but I'm happy that we stayed aligned with our goals. I focused on making sure no feature from our design was left out.

Noe Jr Remolar - Documentation Officer

I spent most of my time capturing the progress and explaining how each page relates to our Use Case. Seeing the project turn from a diagram into a real website is very satisfying. I learned how important it is to have a clear plan before starting to code.

Michael Mantes - Database Manager

My main focus was ensuring that the input fields in the HTML forms matched the data types we set in our MongoDB schemas. If the frontend doesn't match the database, the system won't work, so I double-checked every form field. It helped me understand the connection between the UI and the data.

Nica Alyssa Racadag - FrontEnd Manager

As the Frontend Manager, I handled the actual coding of the pages using Bootstrap 5. I wanted the site to be responsive so that sari-sari store owners could use it on their phones. It was fun experimenting with Bootstrap components like cards and navbars to make the site look modern.

Pear Anne Odo - GitHub Manager

I managed the commits and made sure the /frontend folder was organized properly. I helped the team by coordinating the pushes to GitHub to avoid any code conflicts. I also updated the README file to explain how to navigate the new pages.

NOTE:

AI Usage: *On creating this Phase 3 Lab Report (documentation), around 20% of AI usage was obtained to guide us on finishing this report such as: asking assistance when encountering errors and other suggestions and recommendations that would serve as our guide for improvement. Overall, we hereby acknowledge that all the information and key ideas presented on this phase was based on our actual work, outputs, and understanding of the concepts.*